# NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA
## DEPARTMENT OF ELECTRICAL ENGINEERING
### A.Y 2025-2026 MID Examination– I

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| **Subject** | : | **Problems solving and Programming skills (CSIC103)** | **Batch** | : | **B.Tech 1st sem** |
| **Branches** | : | **EE, SET** | **Time** | : | **50 Minutes** |

**All questions are compulsory** **Max. Marks 15M**

1. In a bank's ATM security system, each customer sets a **4-digit ATM PIN**. For additional fraud detection, the system checks if the **reverse of the entered PIN** matches the original PIN. If the PIN is the same as its reverse (palindrome), the system marks it as a **"Weak PIN"** because it is easy to guess. Otherwise, the PIN is considered a **"Strong PIN"**. **[7M]**

   **Task:** Write a C program that accepts a 4-digit ATM PIN from the user, finds its reverse, and displays both the reversed PIN and whether the PIN is weak or strong based on the comparison.

   Constraints:

   - The PIN must be exactly **4 digits**.
   - If the user enters a number that is not 4 digits, the program should display an error message:
     `"Invalid PIN! Please enter a 4-digit number."`

2. A company gives a **performance bonus** to employees based on their scores in two tests: the **technical test (T)** and the **aptitude test (A)**. If the technical score is greater than 70 and the aptitude score is greater than 60, the employee is awarded a bonus of Rs.**5000**. If this condition is not satisfied but the sum of both scores is greater than 150, the employee is awarded a bonus of Rs.**3000**. Otherwise, the employee is not eligible for a bonus and receives **0**. **[6M]**

   Write a C program that accepts the test scores as input from the keyboard, stores them in variables, and then calculates and displays the bonus based on these conditions using only `if` **statements**.

3. Modify the program in **Question 2** such that: After processing one employee's scores, the program should ask the user if they want to calculate the bonus for another employee. If the user enters **'Y'**, the program repeats or if the user enters **'N' or any other character except 'Y'**, the program should terminate with a thank-you message. **[2M]**

   printf("Enter z m)
   scanf("%d %d ) fool(SY